



ENT-164 · Intro to Making

BEGINNER'S GUIDE · MACOS & WINDOWS · NO EXPERIENCE NEEDED

AI Coding Agent Setup Guide

Accept the class invitation, install the OpenCode app, sign in, and start making.

About 20 minutes

For macOS & Windows

Provider: OpenCode Console

Model: DeepSeek V4.1 Flash

opencode

Ask anything... "Fix broken tests"

Build · DeepSeek V4 Pro Default / OpenCode · max

~

shift+tab agents ctrl+p commands

The OpenCode message box. Type your questions here.

OpenCode Desktop is an AI helper that runs as an app on your laptop. You ask questions in plain English, and it answers them, explains code, and walks you through technical work. That includes the wiring diagrams, the code, and the error messages you can't read.

Figure 1 shows where each part runs. On your laptop: **OpenCode Desktop**, the class Python sandbox that runs the class skills and uploads code to your board, and the ESP32 itself, plugged in over USB. In the cloud: **OpenCode Console**, the service that connects the app to the model with your class account, and **DeepSeek V4.1 Flash**, the model that writes the answers.

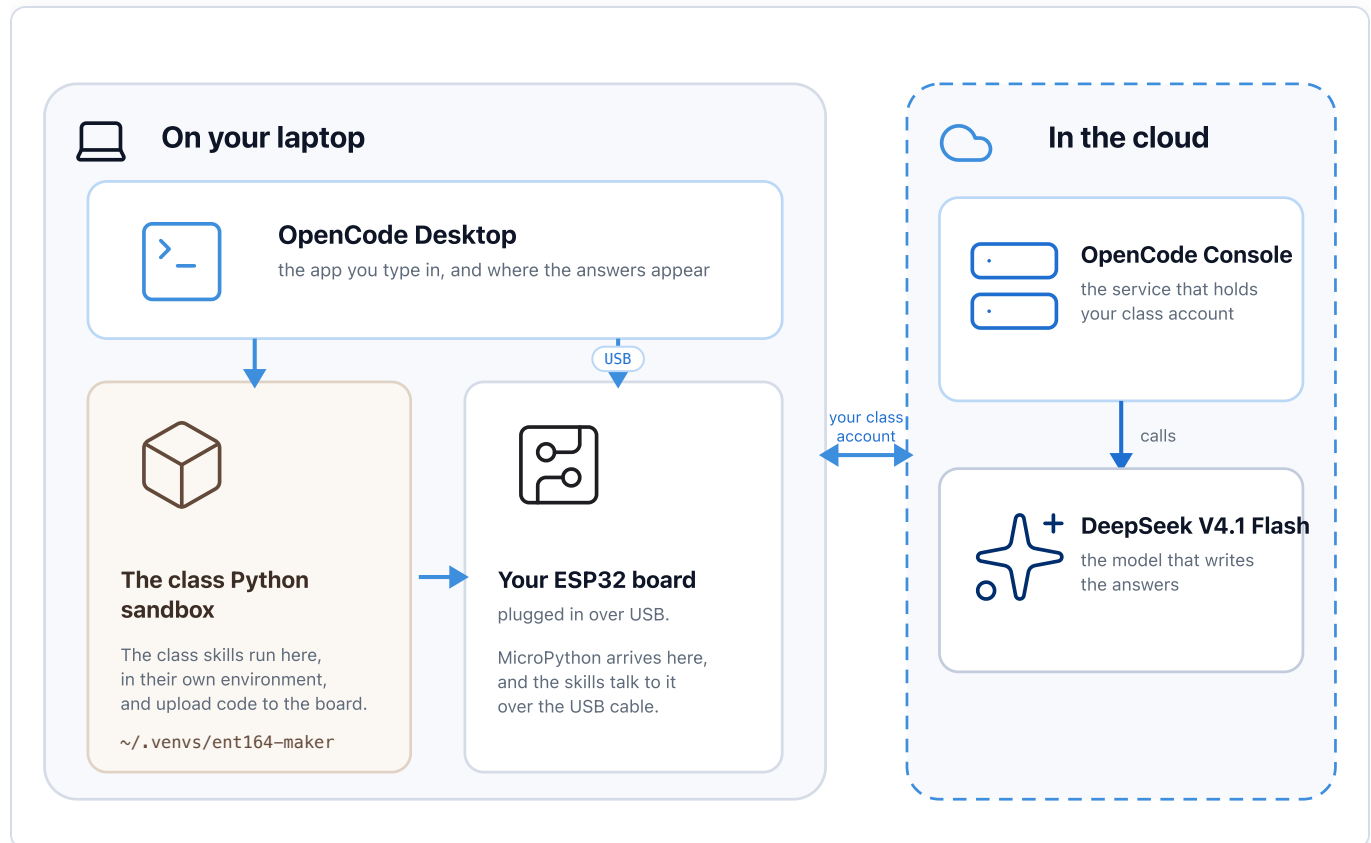


Figure 1. The parts of the system, and where they run. On your laptop: OpenCode Desktop, the Python sandbox that runs the class skills and uploads code to the board, and your ESP32 over USB. In the cloud: OpenCode Console and DeepSeek V4.1 Flash, the model that writes the answers.

What you'll need

- ✓ A laptop running macOS 13 or newer, or Windows 10 or 11
- ✓ An internet connection
- ✓ Your Tufts email. The class invitation goes there, and you sign in with it in Step 1
- ✓ Nothing to pay. The class budget covers AI use for your coursework

1 Accept your class invitation

Your instructor adds you to the class workspace with your **Tufts email**. The invitation arrives by email. You accept it once.

1. Check your Tufts inbox for an invitation to the class workspace and click the link in it.

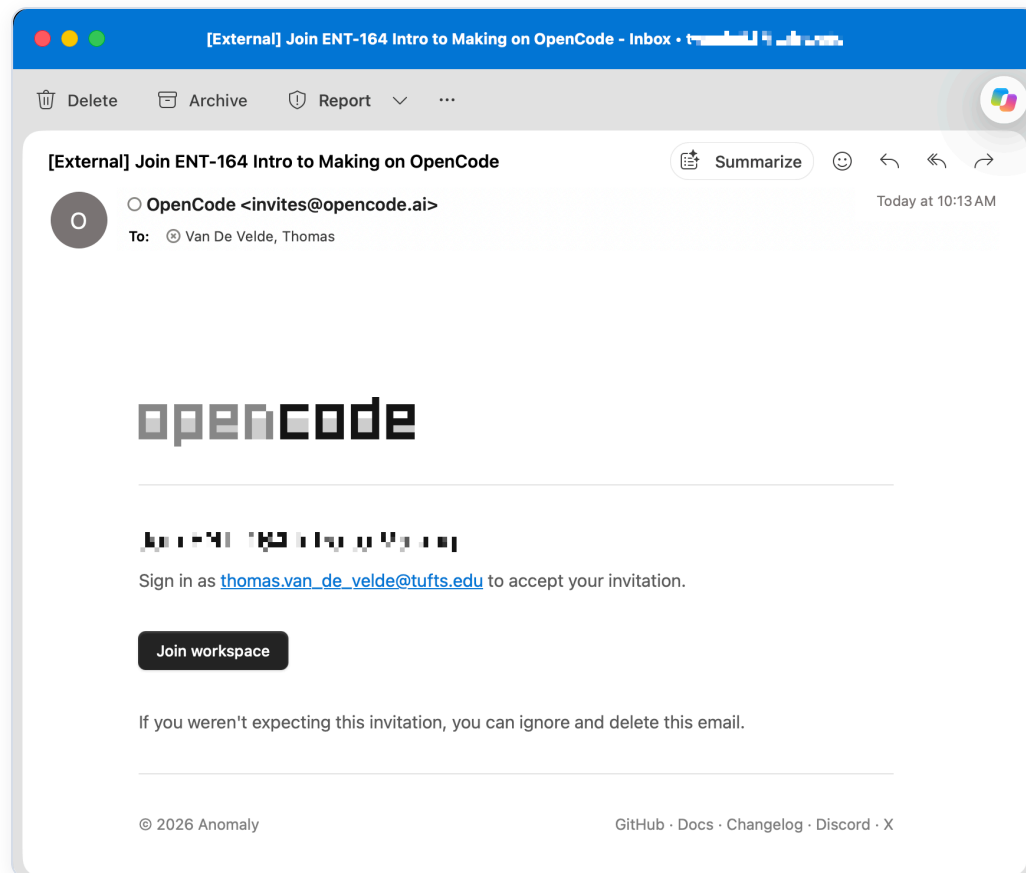


Figure 2. The invitation email. Click **Join workspace** and sign in with the same Tufts address the email was sent to.

2. Click **Continue with Google** and sign in with your Tufts email.

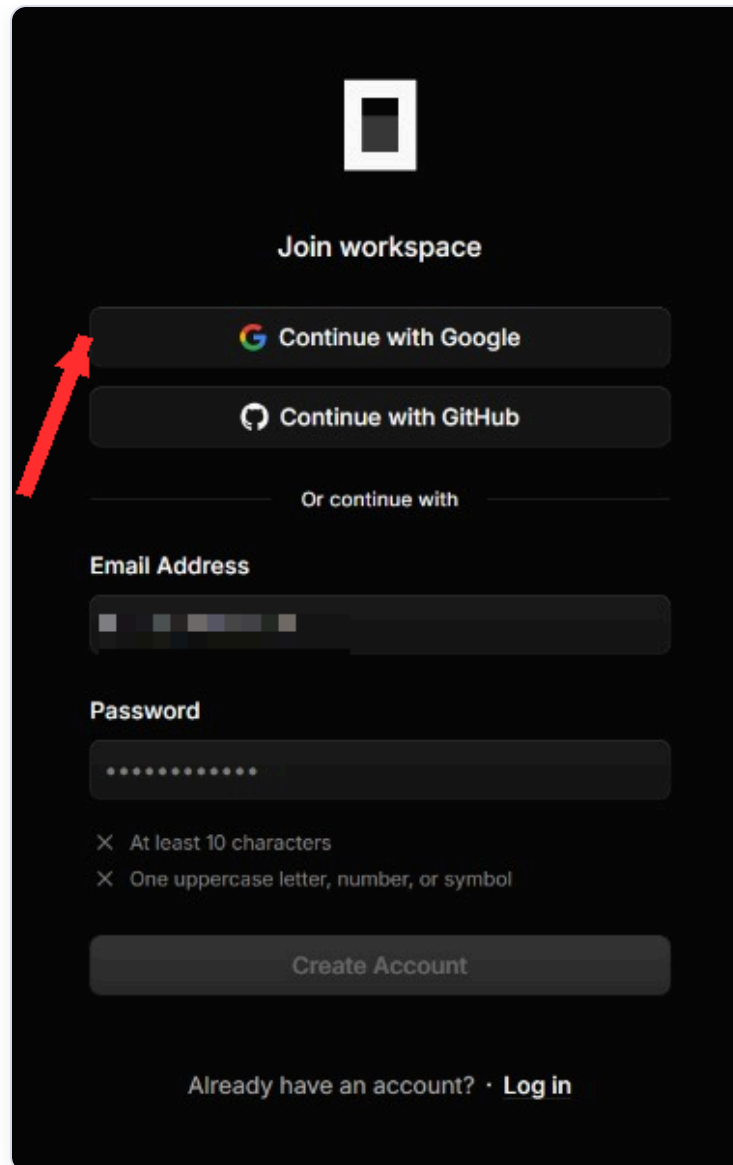


Figure 3. The join page. The red arrow points at **Continue with Google**; sign in with the Tufts address the invitation was sent to.

Use your Tufts email. The invitation is tied to the address your instructor invited. Sign in with anything else and you won't see the class workspace, and the class budget won't apply.

3. You land in the class workspace.

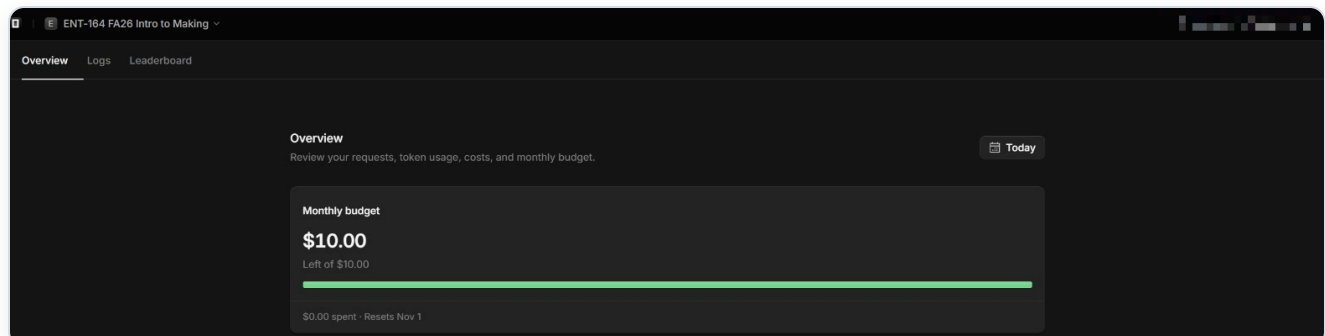


Figure 4. The class workspace in OpenCode Console. The budget card is the class's; your instructor tops it up.

2 Download OpenCode Desktop

1. Open your browser and go to **opencode.ai/download**.
2. Under **OpenCode Desktop**, click the download for your computer: **macOS** (Apple Silicon or Intel) or **Windows (x64)**.
3. The download starts. The file lands in your **Downloads** folder.

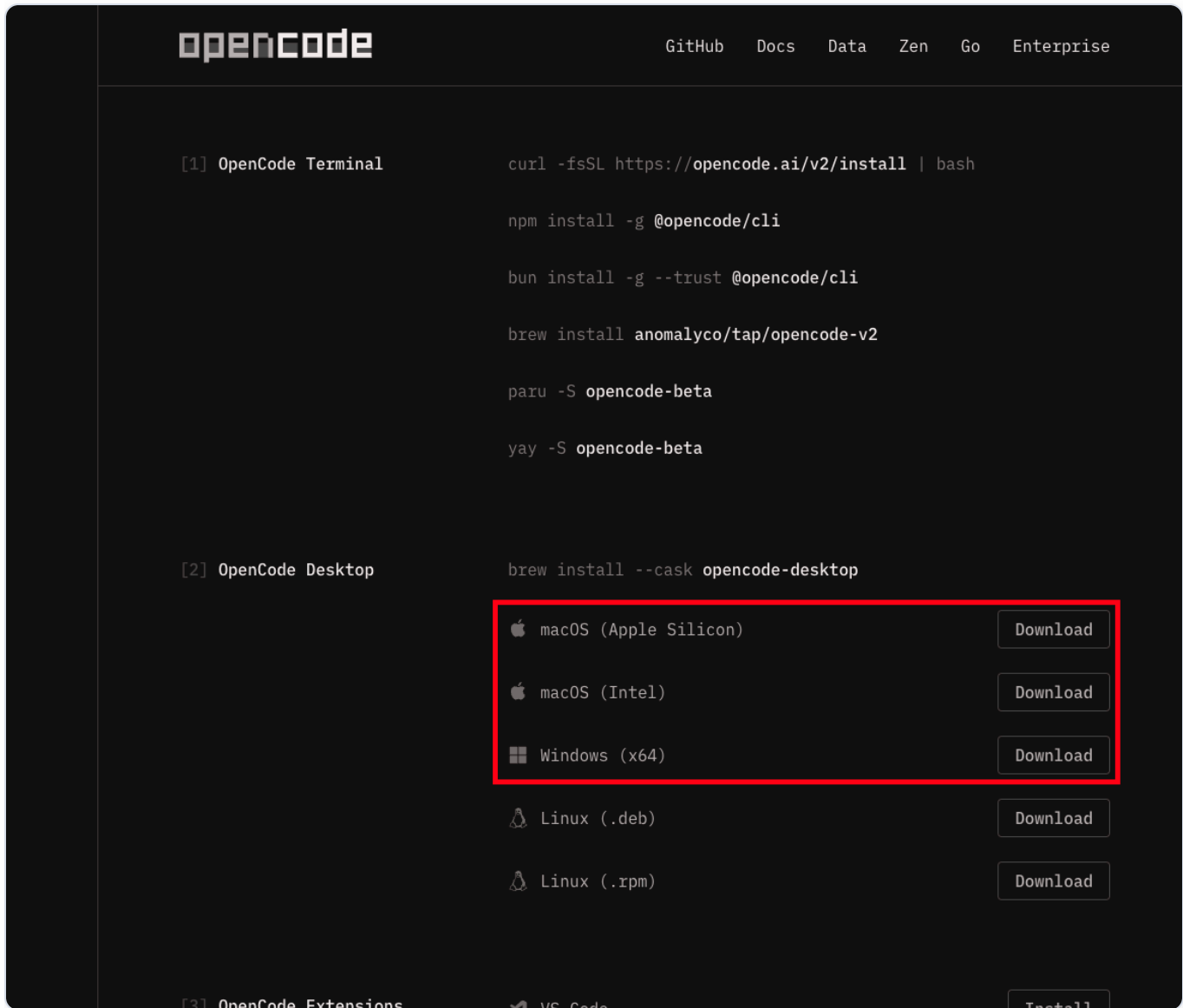


Figure 5. The download page. Under **OpenCode Desktop**, pick the installer for your computer. Both are on the page.

3 Install OpenCode Desktop

On a Mac:

1. Open the downloaded file. Click it in your **Downloads** folder, or in the notification your browser shows.
2. A small window opens showing the **OpenCode** icon and your **Applications** folder.

3. Drag the **OpenCode** icon onto the **Applications** folder, exactly like the arrow shows.
4. Close that window, open **OpenCode** from your Applications folder (or press `⌘ + Space`, type **OpenCode**, and press `Enter`).

On Windows:

1. Open your **Downloads** folder and double-click the OpenCode installer. The file ends in `.exe`.
2. Follow the prompts. If Windows shows "**Windows protected your PC**", click **More info**, then **Run anyway**.
3. When it finishes, open OpenCode from the **Start menu**: press the `Windows` key, type **OpenCode**, and press `Enter`.

If your computer warns about the app. Yes, it looks alarming, and it is the warning every app gets when it arrives from the internet rather than an app store. On a Mac, right-click (Control-click) **OpenCode** in Applications, choose **Open**, and confirm; on Windows, click **More info**, then **Run anyway**. The app is signed and comes straight from `opencode.ai`.

4

Sign in to OpenCode Console

This step connects the app to your class account. You sign in, and that is the whole step: no API key, no password to copy. If a dialog offers you a key, it is offering the wrong thing.

1. Open OpenCode Desktop.
2. Type `/connect` and press `Enter`. A **Connect provider** list opens.
3. Click **OpenCode Console**. It's in the **Popular** section, marked **Recommended**.

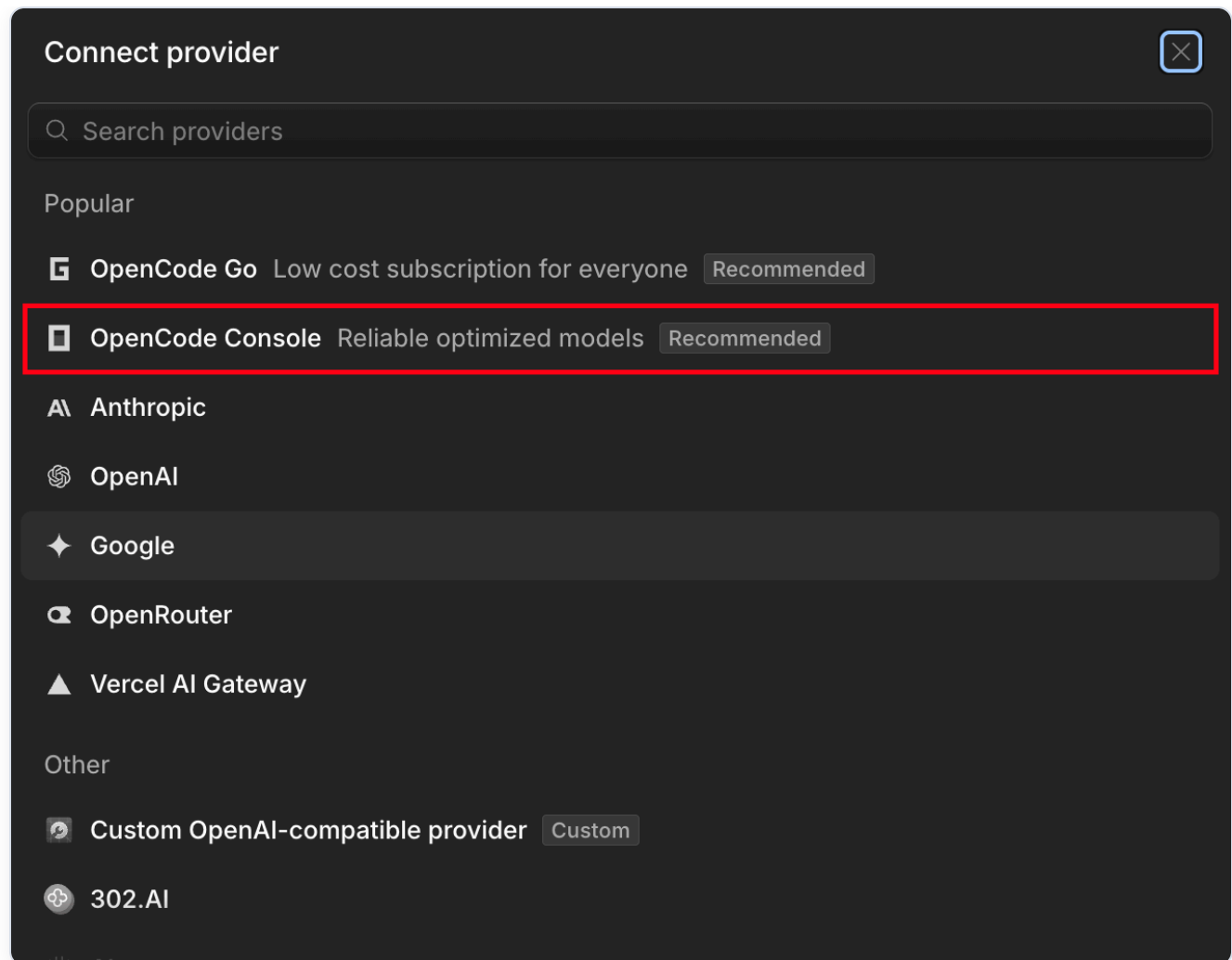


Figure 6. Typing `/connect` opens the provider list. Click **OpenCode Console**, outlined in red.

4. The app opens a browser with a request to authorize OpenCode Desktop. Click the **Authorize** button.

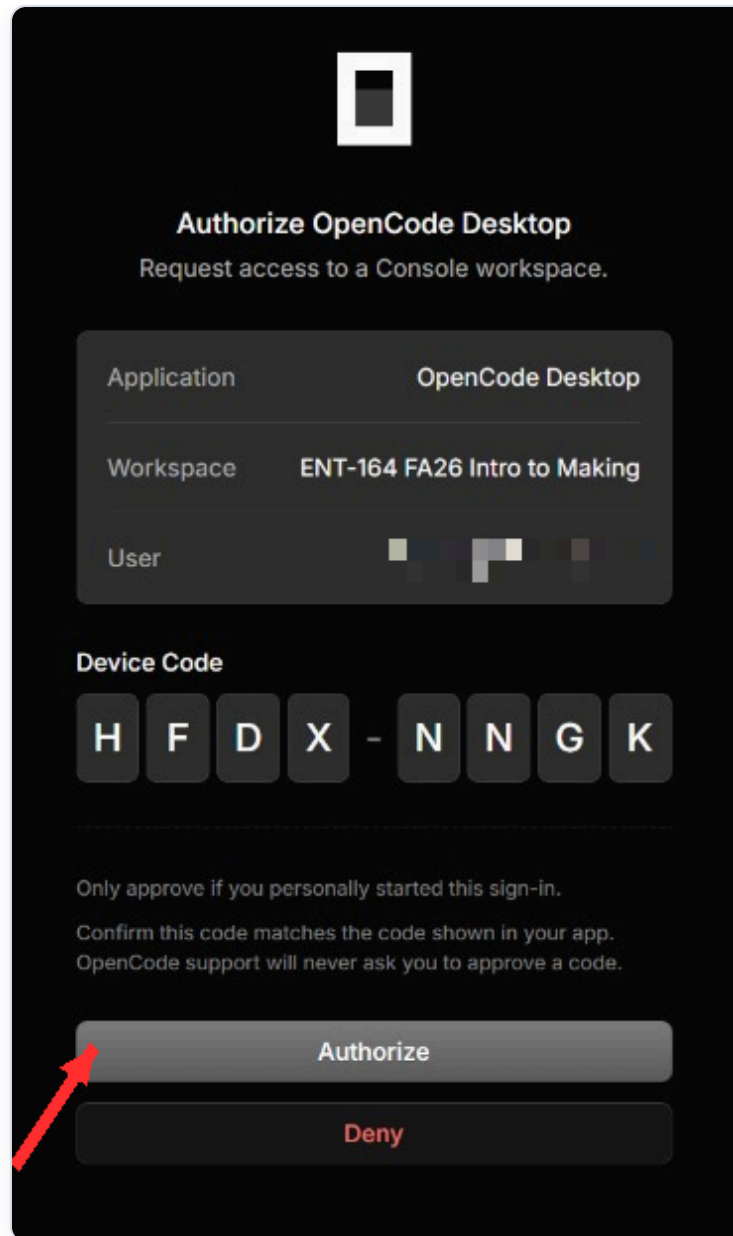


Figure 7. The authorize page. The red arrow marks **Authorize**; click it once the **Workspace** row is the class one and the device code matches the app.

5. Close the browser tab and navigate back to the OpenCode app.

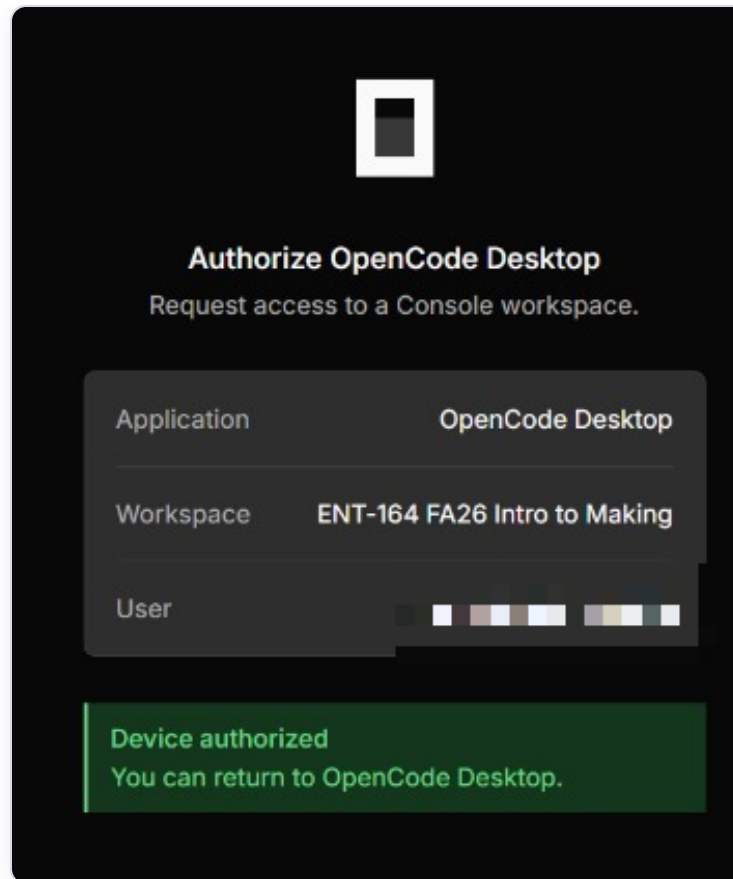


Figure 8. After you approve, the page says **Device authorized**. Close the tab and go back to the app.

6. Verify that **DeepSeek V4.1 Flash** is selected as the model and click the **Continue** button.

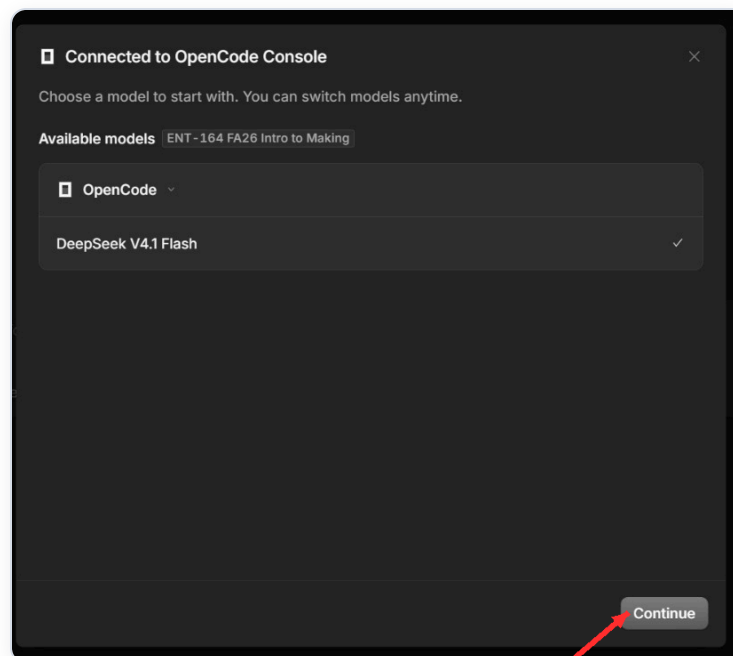


Figure 9. Back in the app: the model picker. **DeepSeek V4.1 Flash** is checked; the red arrow marks **Continue**.

7. You are now ready to start prompting DeepSeek.

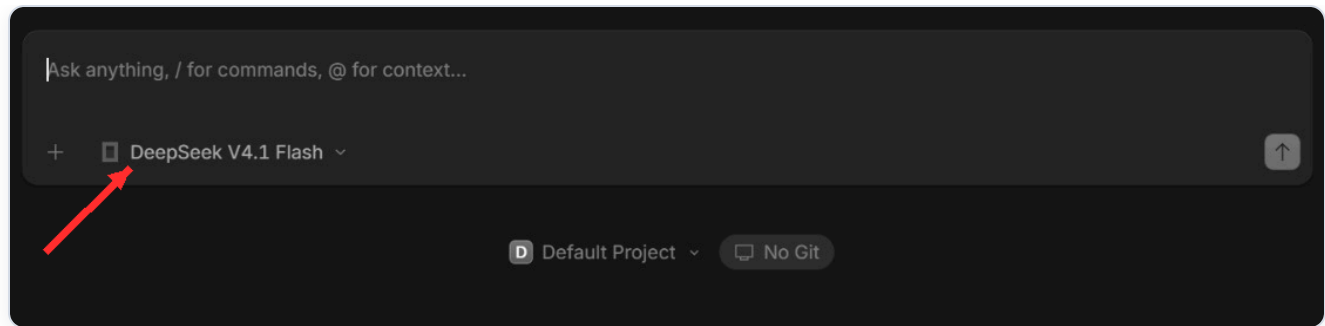


Figure 10. Ready to go. The session runs on **DeepSeek V4.1 Flash**; the red arrow marks it.

5 Ask your first question

1. Click the message box at the bottom that says "Ask anything...".
2. Type a question in plain English, for example: *"I'm new to coding. What can you help me with?"*
3. Press **Enter** (or click the arrow button at the right) and wait a few seconds. The answer appears under your question.

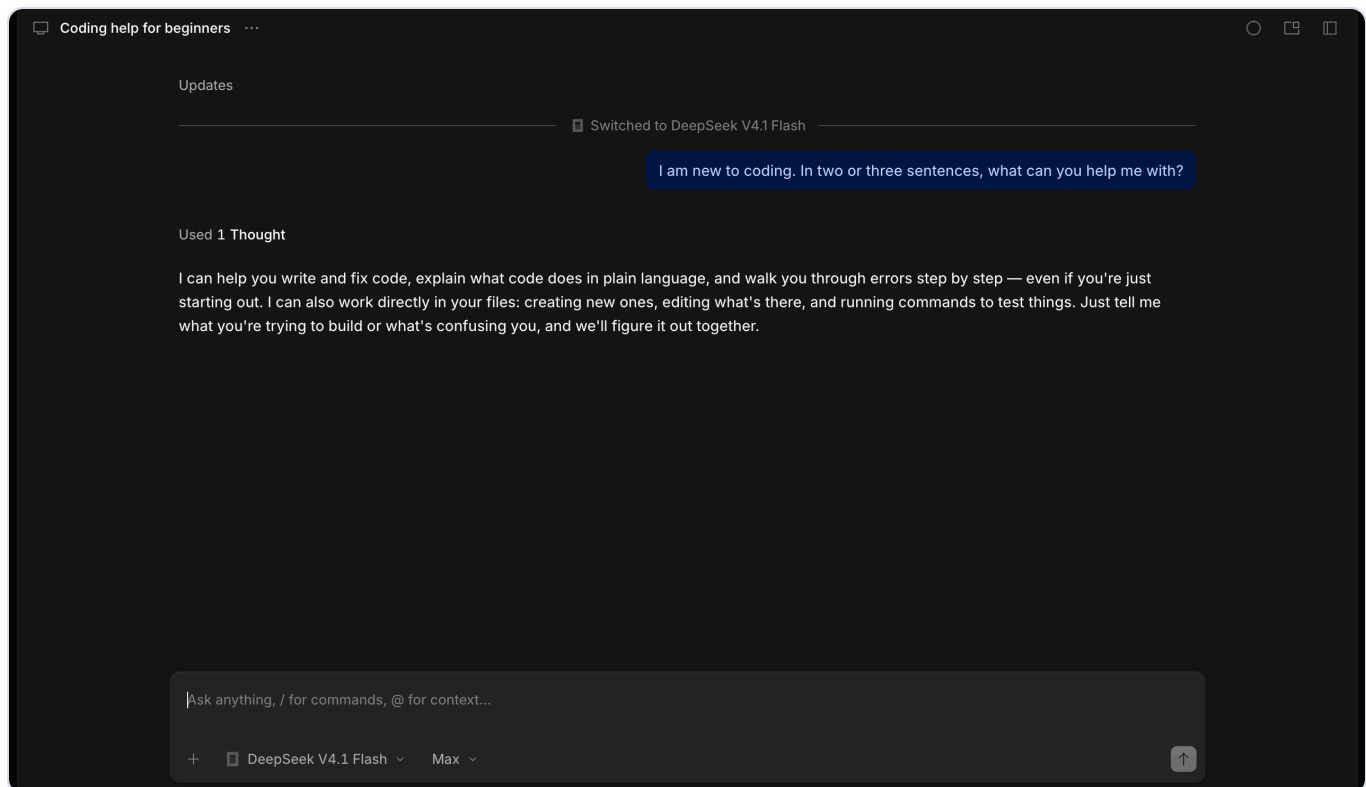


Figure 11. Your question appears in the bubble on the right; the answer gets written below it. The model's name is always visible under the message box.

6 Add your class tools

Finally we'll add class tools to OpenCode. They come as downloadable "skills" hosted on the class website. The skills update automatically when your instructor changes them. The first one you'll try draws

wiring diagrams for your ESP32 kit. Describe a circuit and OpenCode draws a labeled breadboard picture with the jumper wires already routed. It also writes the code and uploads it to your ESP32. Keep the diagram open on the bench while you build.

1. Click in the message box and paste this whole sentence:

COPY AND PASTE THIS

Add my class skills to opencode: <https://tuftsmaker.github.io/skills>

2. Press **Enter** and let it work. It downloads the class tools and remembers where they came from, so it can update them later.
3. If it asks permission, something like "Can I change your settings file?", click **Allow** or **Yes**. It's changing OpenCode's own settings and nothing else on your computer.
4. Quit OpenCode and open it again (on a Mac, **⌘ + Q**; on Windows, close the window). The new tools load at startup.
5. Connect your ESP32 to your laptop using the provided USB cable and enter the following prompt:

COPY AND PASTE THIS

Install MicroPython on my ESP32 and make the internal LED blink.

6. Now try wiring an external LED:

TRY ASKING

Show me how to blink an external LED.

If the answer mentions the class tools and the wiring diagrams, you're set. Ask for a diagram whenever you need one, for example "Draw me a wiring diagram for connecting an LED to my ESP32". Each one is saved in your project folder as a picture you can zoom and trace.

One-time step. The class tools update themselves when your instructor improves them. You never repeat this.

If something goes wrong

WHAT YOU SEE	WHAT TO DO
Your computer won't open the app	Mac: in Applications, right-click (Control-click) OpenCode and choose Open , or use System Settings → Privacy & Security → Open Anyway . Windows: in the installer warning, click More info → Run anyway .

WHAT YOU SEE	WHAT TO DO
No invitation email	Check your Tufts inbox's spam folder, and confirm your instructor has your correct <code>tufts.edu</code> address. Your instructor can send it again.
The invitation page says it belongs to a different account	Your browser is signed in as someone else. Click Sign out and continue on that page, then sign in with the Tufts address the invitation was sent to.
Can't find the provider panel	Type /connect in the message box. The provider list opens there. The same list is under Settings → Providers . If Settings shows a Servers list, open your server first (usually Local Server).
OpenCode Console isn't in the list	It's in Popular providers near the top, marked Recommended . If you still don't see it, click Show more providers .
The browser doesn't open for sign-in	Click Copy sign-in link in the app, then paste it into your browser's address bar.
"This sign-in request has expired"	The device code timed out. Click Try again to get a fresh one, and finish the sign-in a little faster.
Signed in but DeepSeek isn't in the model list	You may have authorized a different workspace. Go back to Settings → Providers , Disconnect OpenCode Console, connect again, and this time switch the workspace to ENT-164 Intro to Making on the authorization page. If it still doesn't appear, use Manage models in the model panel, search v4.1 , and switch it on.
An error about credits or balance	Don't add credit yourself. Check that you signed in to the ENT-164 Intro to Making workspace, then tell your instructor. The shared budget may be out of funds.
It asks for an API key	You don't need one. Close that dialog and use the account sign-in instead; the API key option is only for shared service accounts.
OpenCode says it can't reach the website (Step 6)	Check your wifi. The class tools download from the class website the first time.
The class tools don't seem to be on	Ask OpenCode <i>"Show me how to blink an external LED."</i> If it doesn't know the class tools, paste the sentence from Step 6 again, then quit and reopen OpenCode.
No answer, or it seems stuck	Wait ten seconds. The first reply can be slow. Check your internet, then ask again. Quitting and reopening the app also helps.

Words you'll see

OpenCode	The app on your computer where you chat with the AI.
OpenCode Console	The service that connects the app to a tested set of AI models. You reach it by signing in with your class account.
Device code	The short code the app shows during sign-in. Your browser shows the same code, and you approve it there.
Workspace	The shared class area in OpenCode Console. Here, ENT-164 Intro to Making . Choose it when the sign-in page asks.
Class tools (skills)	Extra abilities the class website teaches OpenCode, including the wiring-diagram maker. OpenCode downloads its own copy and keeps it up to date.
Provider	The service that runs the AI connection. Here, OpenCode Console.
Model	The AI model you're talking to. Here, DeepSeek V4.1 Flash.
Session / tab	One conversation. Each tab at the top of the app is a separate chat.

Where everything lives

WHAT YOU WANT	WHERE TO CLICK
Accept the class invitation	The link in the email your instructor sent to your Tufts address
Sign in or manage your account	opencode.ai/console . Sign in with your Tufts email.
Connect or fix your provider	Type /connect in the message box → OpenCode Console (or Settings → Providers ; if Settings shows a Servers list, open your server first (usually Local Server))
Switch to the class workspace	On the authorization page during sign-in, choose ENT-164 Intro to Making
Start a new chat	The + next to the tabs at the very top
Change the model	The model button under the message box (shows "DeepSeek V4.1 Flash")
Show all models	Model button → Manage models
Your class tools and wiring diagrams	The class tools live at tuftsmaker.github.io/skills ; every diagram you make is saved in your project folder as a picture
Quit the app	Mac:  + Q · Windows: close the window or  + F4

Things to try in your first week

You don't need to know any code to start. Ask in plain English:

YOU CAN ASK	WHAT YOU GET
<i>"I'm new to coding. What can you help me with?"</i>	A plain-language list of what it can do, and where to start.
<i>"Explain this like I've never seen it before"</i>	Any file or concept broken down step by step, in beginner terms.
<i>"Draw me a wiring diagram for connecting an LED to my ESP32"</i>	A labelled breadboard picture, once the class tools are on (Step 6).
<i>"Write the code for a button that lights an LED"</i>	The MicroPython code, with the wiring it needs.

Once you're comfortable, the official docs at [opencode.ai/docs](#) show what else OpenCode can do.